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運用大型語言模型分析多元金融文本於股票選擇策略
Stock Selection Strategies through the Analysis of Diverse
Financial Texts Using Large Language Models

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摘 要

近年，大型語言模型在自然語言處理領域展現出顯著的突破與進步。研究運用提示工程，設計多樣化的指令提示，讓模型有效解析來自新聞、財報與總體經濟事件等多元金融文本，並產出具語意結構的標籤與判斷，作為股票選擇策略的基礎信號來源。針對三類金融文本分別設計提示策略：對新聞標題進行情緒分類；對財報與總經類文本則採用鏈式思考提示，引導模型推理語境與潛在市場展望。整體方法採用零樣本學習設計，讓模型能直接執行分類與推論任務。

語言模型產生的語意判斷結果進一步轉換為結構化特徵，供機器學習模型使用，以驗證其對股價變動的預測能力，以評估語意特徵的資訊價值。基於語意門檻的規則式多空策略則進行回測，驗證其在實際應用的可行性。實驗結果顯示，語意特徵具備良好預測能力，規則策略亦能穩健產生超額報酬。本研究證實語言模型結合提示設計與多元金融文本，能有效擷取具操作性的語意訊號，提升量化選股策略的應用性，並為語意驅動選股與智慧投資提供實證基礎。

關鍵字：大型語言模型、提示工程、鏈式思考、金融文本分析、股價預測

Stock Selection Strategies through the Analysis of Diverse Financial Texts Using Large Language Models

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Abstract

Over the past few years, large language models (LLMs) have achieved remarkable progress in natural language processing. This study applies prompt engineering to guide LLMs in analyzing financial texts, including news, earnings reports, and macroeconomic content. The model generates semantic labels and judgments as signals for stock selection strategies. Prompts are tailored to each text type, using sentiment classification for news and chain-of-thought reasoning for narrative texts to infer market outlook. The approach employs zero-shot learning, allowing the model to perform tasks directly based on prompts.

Semantic outputs are transformed into structured features and evaluated through a machine learning model to verify their predictive value. A rule-based long-short strategy using semantic thresholds is backtested to assess practical effectiveness. Results show that semantic features provide meaningful predictive power, and the rule-based strategy delivers stable excess returns. This study confirms that language models can extract actionable semantic signals from financial texts and support their application in quantitative stock selection.

Keywords: Large Language Models, Prompt Engineering, Chain-of-Thought Prompts , Financial Text Analysis, Stock Price Prediction

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Chapter 1. Introduction

1.1 Background

Designing effective stock selection strategies is inherently difficult because financial markets are complex, highly volatile, and influenced by multiple interrelated factors, as evidenced by market anomalies like the momentum effect [1]. Investors must consider various heterogeneous data types. Traditional models rely on quantitative inputs like historical prices and technical indicators, but these alone fail to capture investor sentiment or macroeconomic shocks. To address this gap, incorporating textual data provides a fuller view of market dynamics. Financial news captures short-term sentiment, often influencing immediate price movements [2]. Earnings reports offer structured assessments of a company's financial health and management outlook, useful for evaluating long-term prospects [3]. Macroeconomic statements signal policy direction and economic momentum at broader levels [4]. Collectively, these three text types provide complementary insights essential to more informed and robust stock selection strategies.

The analysis of financial texts remains a challenging task in investment research. News articles often contain implicit semantic cues and are expressed in diverse ways, making their meanings difficult to interpret explicitly. This challenge is further underscored by the limitations of traditional text analysis methods in capturing subtle variations and contextual logic[5][6]. Earnings reports and macroeconomic texts are based on structured data. However, they are often written in neutral or descriptive language and require further interpretation to understand their potential impact on the market. Traditional

keyword-based sentiment tools and simple classification models often fall short in capturing the subtle semantics and contextual logic embedded in financial texts[7]. These limitations underscore the need for more advanced techniques to effectively extract meaningful information for more robust stock selection strategies.

Recent developments in Natural Language Processing (NLP) have enabled more effective tools for understanding unstructured financial data. Among them, Large Language Models (LLMs) show strong capabilities in capturing semantics, modeling contextual dependencies, and reasoning over complex text [8][9]. These models are particularly well-suited for analyzing financial narratives such as news articles, earnings reports, and macroeconomic statements, which often require interpretation beyond surface-level patterns. Empirical studies have demonstrated the effectiveness of LLMs in addressing various financial tasks. Kirtac and Germano found that LLMs outperformed traditional dictionary-based methods, such as the Loughran – McDonald lexicon, in detecting sentiment in financial news[10]. Similarly, Kim et al. showed that LLMs could predict earnings changes using anonymized financial statements, outperforming analysts and conventional models[11]. LLMs are increasingly being adopted in financial practice. According to *The Financial Times*, both institutional and retail investors are integrating them into investment research and advisory workflows, reflecting their growing role in real-world decision-making[12].

Although LLMs possess strong semantic and reasoning capabilities, they are prone to hallucination and are highly sensitive to input formatting and task framing. To improve their accuracy and stability when interpreting heterogeneous financial texts, this study introduces a prompt engineering approach. By designing structured, task-specific prompts, the model is guided to perform semantic processing and reasoning tailored to different content types [13]. The approach applies basic sentiment classification to news headlines

and uses Chain-of-Thought prompts for advanced reasoning on descriptive content such as earnings commentaries and macroeconomic reports[14]. This strategy allows for differentiated semantic processing across varied text sources and supports more nuanced interpretations of financial information. This design not only improves semantic accuracy and consistency, but also enables the integration of diverse financial signals at the feature level. This study provides a systematic framework for applying prompt-based reasoning to financial texts and advances the development of practical forecasting tools for stock markets.

1.2 Motivation and Research Problems

Although Large Language Models (LLMs) demonstrate strong proficiency in interpreting financial texts, fully leveraging their potential for reasoning and analysis across diverse financial narratives remains a challenge. News headlines tend to convey sentiment and short-term market signals, whereas earnings reports and macroeconomic statements contain descriptive content and require deeper contextual reasoning. If LLMs aren't specifically guided to apply appropriate reasoning pathways for each document type, their ability to extract critical, context-aware insights for investment tasks can be significantly limited[15].

To address this, this study adopts a prompt engineering strategy that aligns each financial document type with a task-specific prompting structure. Sentiment classification is employed for news headlines, while Chain-of-Thought (CoT) prompts enable step-by-step reasoning in more descriptive content such as earnings reports and macroeconomic statements. This approach facilitates differentiated semantic extraction, which can then be translated into actionable insights. Based on this motivation, the study explores the following research questions:

(1) Can task-specific prompts help LLMs generate responses suitable for sentiment analysis and reasoning across different financial texts?

This study investigates whether prompts tailored to specific financial tasks can guide LLMs to produce responses that are more focused, interpretable, and actionable. To enable this, structured data such as earnings reports and macroeconomic indicators are converted into textual summaries, allowing LLMs to process them as natural language. Although these texts are relatively simple and descriptive in form, they often contain implicit signals about economic trends, policy direction, and market expectations that cannot be easily extracted using surface-level methods.

To interpret such signals, this study utilizes Chain-of-Thought prompting to lead LLMs through a sequential reasoning process when analyzing earnings and macroeconomic texts. Rather than merely restating the data, the model is encouraged to incorporate contextual knowledge and historical patterns to infer potential market implications. In contrast, sentiment classification is used for news headlines to capture short-term market sentiment. Through this differentiated strategy, the model can adjust its processing based on the type of financial text and produce insights relevant to investment decision-making.

(2) Can the features extracted from prompt-guided responses enhance stock prediction and improve stock selection strategies?

This study further investigates whether semantic features extracted from prompt-guided LLM responses can enhance the accuracy of stock return prediction and improve stock selection strategies. These features, such as sentiment and forward-looking, are derived from the model's interpretation and reasoning over financial texts, including news articles, earnings reports, and macroeconomic statements.

Through task-specific prompt design, the model is guided not only to identify the

emotional tone of the text but also to infer its potential market implications. The extracted semantic signals are then quantified into high-level features and incorporated into forecasting and selection models. This approach is expected to complement traditional financial and technical indicators, thereby enhancing both the interpretability and robustness of predictive models in real-world investment applications.

1.3 Research Objectives and Contributions

This study aims to use prompt engineering to guide LLMs in processing various types of financial texts, extracting semantically meaningful and actionable insights for stock return prediction and selection.

The research objectives are threefold: (1) to design prompt formats that match the characteristics of each text type, using sentiment classification for news and applying chain-of-thought reasoning for earnings and macroeconomic reports; (2) to convert LLM outputs into semantic features such as sentiment scores and trend indicators, and integrate them with technical signals to create multimodal inputs; and (3) to assess the utility of these features through machine learning – based prediction models and rule-based long-short investment strategies.

This study contributes a modular framework for semantic signal extraction that requires no additional training and is adaptable across financial text domains. It demonstrates that LLM-derived features improve stock prediction accuracy and further shows, through backtesting, their robustness and excess return potential in practical investment strategies.

1.4 Thesis Outline

The remainder of this essay is structured as follows: Chapter 2 provides a review of the related literature. Chapter 3 outlines the proposed system framework. Chapter 4 details the experimental setup, while Chapter 5 discusses the evaluation results and key findings. Finally, Chapter 6 concludes the study by summarizing its contributions, acknowledging limitations, and suggesting directions for future research.

Chapter 2. Related Literature

2.1 Large Language Models in Natural Language Processing

With the rapid development of Natural Language Processing (NLP) technologies, LLMs have increasingly become a fundamental element in tasks involving language understanding and generation. Since the introduction of the Transformer architecture by Vaswani et al., its self-attention mechanism has demonstrated strong capabilities in capturing long-range semantic dependencies, thereby significantly enhancing the effectiveness of language modeling[16]. Building on the Transformer architecture, models such as BERT [17], the GPT series [8], T5 [18], and LLaMA [9] have become foundational tools in NLP, powering tasks such as categorizing text, answering questions, generating dialogues, and producing summaries.

The strong performance of these models across diverse tasks can be attributed to their semantic representation capabilities. Compared to traditional word embedding methods such as Word2Vec and GloVe, the semantic vectors generated by LLMs are context-aware word representations that dynamically adjust to sentence-level context, effectively mitigating issues of word sense ambiguity and semantic vagueness. As a result, LLMs demonstrate clear advantages in tasks such as semantic understanding, natural language inference, and sentiment analysis, and have increasingly become essential technologies in cross-lingual and cross-domain applications.

Moreover, LLMs have demonstrated strong performance in zero-shot and few-shot learning tasks. Models such as GPT-3, for instance, can perform a wide range of downstream

tasks with only a few prompt examples, highlighting their ability to generalize across a wide range of language tasks[8].

2.2 Large Language Models for Financial Text Analysis

LLMs have become a key technology in financial text analysis, owing to their outstanding abilities in semantic comprehension and contextual interpretation. Financial information is predominantly conveyed through vast amounts of unstructured text, such as news reports, financial statement commentaries, and macroeconomic releases. These narratives are rich in complex semantic cues, including implicit sentiment, corporate outlooks, and critical policy signals. Traditional analytical methods, like rule-based classifiers or fixed sentiment dictionaries, often struggle to fully capture these nuanced and implicit meanings[19, 20]. To address the domain-specific challenges of financial sentiment analysis, Araci (2019) introduced FinBERT, a BERT-based model pre-trained on financial corpora. FinBERT demonstrated significant improvements over dictionary-based approaches such as the Loughran–McDonald lexicon[21].

Recent empirical studies have consistently demonstrated the advanced capabilities of LLMs, such as GPT-4 and LLaMA models, across a wide spectrum of financial tasks, significantly outperforming conventional approaches. For instance, in financial news sentiment analysis, Kirtac and Germano found that large language models generally outperformed traditional dictionary-based methods, such as the Loughran–McDonald lexicon, in detecting sentiment [10]. Extending this line of evidence, Lopez-Lira and Tang [22] showed that LLM-derived sentiment from financial news can predict next-day stock returns, demonstrating the model’s effectiveness in real-world financial forecasting. Beyond news, LLMs have also exhibited strong promise in sophisticated corporate financial report analysis, with

studies confirming their ability to infer earnings changes from anonymized financial statements, demonstrating their comprehension of complex financial discourse and capacity to support intricate decision-making[11]. This trend is exemplified by BloombergGPT, a 50-billion-parameter model trained on a vast mix of financial and general-domain data, which outperforms general-purpose LLMs on a wide range of financial NLP tasks[23].

Crucially for broader market insights, LLMs have shown remarkable proficiency in macroeconomic text analysis and economic reasoning. Specifically, for central bank communications, Kim et al. (2023) investigated LLMs' effectiveness in analyzing FOMC minutes, highlighting their ability to capture nuanced economic stances, though also noting existing constraints [24]. Extending this, Guo and Yang (2024) systematically evaluated LLMs on complex economic reasoning tasks, demonstrating their capacity to infer causal relationships and understand economic scenarios, though limitations persist in terms of consistency, factual grounding, and handling of subtle economic intricacies [25]. Furthermore, LLMs are being actively explored for macroeconomic forecasting [26] and for extracting crucial economic signals from unstructured data, such as government statements [27].

The effectiveness of LLMs in financial text analysis is largely attributable to their Transformer-based architecture. This architecture equips LLMs with the capability to effectively capture long-range contextual dependencies and parse the multi-layered semantic structures inherent in financial language, including specialized terminology, subtle tonal shifts, and complex context dependencies. This enables LLMs to go beyond basic sentiment classification, performing more intricate reasoning and inference tasks, such as identifying key operational strategies from corporate financial reports, forecasting policy shifts from macroeconomic reports, and predicting short-term volatility from market news. Through

these deeper semantic understanding capabilities, LLMs offer the potential to significantly enhance the precision and comprehensiveness of financial analysis.

Overall, the application of LLMs in financial text analysis is opening up new possibilities. They show promise in significantly boosting the efficiency and depth of processing large-scale unstructured data, enabling the automatic identification and integration of crucial information dispersed across various textual sources. This can help investors and analysts gain a more comprehensive understanding of market dynamics and potentially uncover hidden patterns that traditional methods might miss. Whether leveraging their general pre-trained capabilities, fine-tuning them for specific financial sub-domains, or utilizing sophisticated prompt engineering designs [13], LLMs can provide robust semantic analysis support for financial decision-making, positioning them as a noteworthy tool in modern financial analysis.

2.3 Prompt Engineering for Financial Language Understanding

Despite their remarkable proficiency in language comprehension and generation, large language models (LLMs) often encounter challenges when used for specialized tasks in domains like financial analysis. Fine-tuning approaches, though effective, typically require large labeled datasets, significant computational resources, and task-specific retraining. These challenges make them less practical in financial contexts, where annotated data is scarce and text patterns vary widely across document types such as news headlines, earnings summaries, and macroeconomic reports [13].

Prompt engineering offers a lightweight and scalable alternative to fine-tuning. Instead of modifying model parameters, this approach guides LLMs to perform specific tasks through natural language instructions that clearly and contextually define the task. With

carefully crafted prompts, LLMs can adjust their responses according to the distinct semantic and structural characteristics of different financial texts, even under zero-shot conditions. For example, in the MarketSenseAI system, researchers employed Chain-of-Thought reasoning and In-Context Learning techniques to guide GPT-4 through prompt engineering for financial data analysis and stock selection. Their results show that even without additional fine-tuning, carefully designed prompts can yield highly effective investment recommendations[28].

Recent studies have demonstrated that well-designed prompts can elicit structured and informative outputs from LLMs without the need for additional training. These outputs may include sentiment assessments[29], interpretations of economic or policy implications, and evaluations of corporate outlooks[11]. Notably, the FinGPT and Instruct-FinGPT frameworks have shown that instruction-tuned or prompt-optimized LLMs can surpass both traditional models and even general-purpose LLMs in financial sentiment tasks, especially when dealing with numerical sensitivity and ambiguous context [30]. Prompt engineering methods can vary in form, such as those that use direct task descriptions to instruct the model, others that guide the model to reason through problems in a step-by-step manner (e.g., Chain-of-Thought prompting)[14], and those that include a small number of examples to illustrate the expected response (e.g., few-shot learning).

In financial applications, prompt engineering has shown particular promise in scenarios with limited labeled data. When prompt formats are adapted to the characteristics of different textual sources, models are more likely to produce accurate and semantically rich outputs. For example, directive prompts are effective for concise news headlines, while more interpretive prompts can support reasoning in longer and more descriptive financial documents. This strategy enables the extraction of meaningful semantic signals

from heterogeneous data sources without extensive model retraining, offering a practical and cost-efficient solution for real-world financial forecasting and decision-making tasks.

2.4 Behavioral Finance and Market Efficiency

With the advancement of data processing and natural language processing (NLP) technologies, financial theory has continued to evolve in its understanding of financial behavior, offering more practically grounded analytical perspectives. The conventional Efficient Market Hypothesis (EMH) claims that all publicly available information is immediately and completely incorporated into asset prices.

However, recent developments in behavioral finance suggest that markets may react inefficiently to emotional and semantic signals embedded in textual data, exhibiting short-term volatility spikes followed by delayed price corrections. For example, Tetlock finds that a pessimistic media tone predicts short-term stock price declines, which are typically reversed in subsequent days—highlighting systematic inefficiencies in how markets process sentiment-laden information[31]. These findings align with a broader body of behavioral finance research showing that investor behavior is often driven by cognitive biases such as overconfidence and herd mentality, which can lead to mispricing and sentiment-driven anomalies in asset returns [32, 33]. Such evidence suggests that when the market engages with unstructured information, particularly semantic and emotional signals, significant inefficiencies can arise, which challenges the foundational assumptions of the EMH.

Prior studies show that meaning and emotion in financial texts can affect asset prices. Traditional models often struggle to capture these features, while recent advances suggest that LLMs improve sentiment extraction and return prediction. For example, Kirtac and Germano (2024) investigated how financial news sentiment relates to stock returns in the

U.S., demonstrating that sentiment scores generated by the OPT model exhibited strong predictive capabilities, highlighting the potential of LLMs to advance the integration of behavioral finance and machine learning[10].

Building on this perspective, this study adopts the Efficient Market Hypothesis (EMH) as a theoretical benchmark and draws on the behavioral finance view of market irrationality as its motivation. This study employs language models to capture semantic content embedded in financial language and investigates how semantic shifts in financial text relate to asset price movements over time, aiming to uncover systematic biases in market interpretation.

Chapter 3. System Framework

In this study, we propose a stock selection mechanism that leverages semantic signals extracted from large language models (LLMs) via prompt engineering. Task-specific prompts are designed for different categories of financial texts, including news headlines, earnings report summaries, and macroeconomic event descriptions. These prompts guide LLMs to perform zero-shot semantic inference for each type of input. The outputs, such as sentiment classification and market outlook assessments, are processed within the LLM-Derived Signal Extraction Module to generate structured features. These features are then combined with traditional technical indicators to improve the performance and interpretability of downstream return prediction models. Ultimately, these semantic signals serve as the basis for stock selection decisions.

The process of this mechanism is as follows:

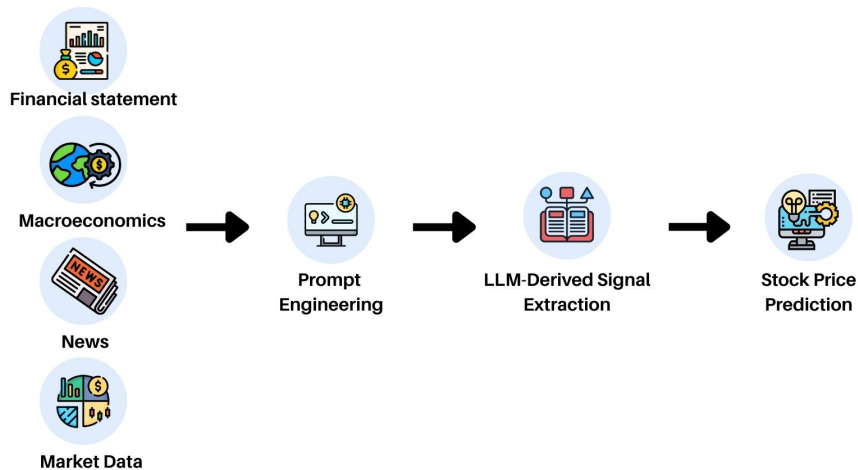


Figure 1: Process Flow

1. We collect and preprocess financial data using the EODHD API. The dataset encompasses four key perspectives: stock prices, financial news, company fundamentals, and macroeconomic narratives. This integrated dataset bridges structured and unstructured financial information, serving as the foundation for downstream prompt-based inference and predictive modeling.

2. Next, we then construct task-specific prompts to guide large language models (LLMs) in interpreting three categories of financial texts: news headlines, financial statement summaries, and macroeconomic descriptions. Each prompt is tailored to its text type—targeting sentiment classification for news, corporate outlook reasoning for earnings, and economic sentiment assessment for macro narratives. This design enables semantic understanding across heterogeneous financial documents through zero-shot prompting.

3. Based on the designed prompts, LLMs generate semantic outputs such as sentiment polarity and market tone directly from financial texts. These outputs are subsequently processed in the LLM-Derived Signal Structuring Module, where they are converted into structured numerical features, including sentiment scores, trend dynamics, and signal volatility. These semantic signals form part of the input for downstream prediction models.

4. In the final stage, we integrate LLM-driven semantic signals with traditional technical indicators to forecast stock price movements using machine learning and deep learning models. Predictive performance is evaluated using standard regression metrics such as Mean Squared Error (MSE) and the coefficient of determination (R^2). Furthermore, we conduct signal-based backtesting to evaluate the effectiveness of LLM-driven features in real-world portfolio strategies, providing practical insights into their financial applicability.

Our system consists of four modules, which are listed as follows:

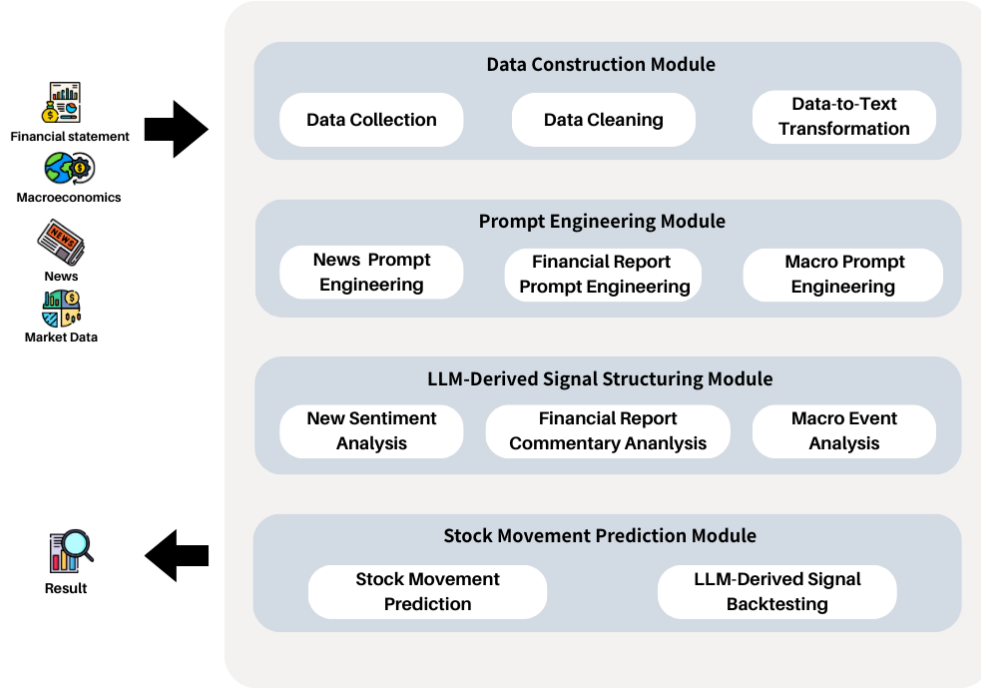


Figure 2: System Framework

(1) **Data Construction Module:** In this module, the first step is data collection. We gather financial data through the EODHD API, including stock prices, financial news articles, company fundamentals, and macroeconomic event descriptions. The second step involves data cleaning, which removes abnormal characters, duplicate or highly similar news articles, and other types of noise. The third step is data-to-text transformation, which converts structured data, such as financial statements and economic indicators, into natural language descriptions suitable for processing by large language models.

(2) **Prompt Engineering Module:** This module focuses on the design of task-specific prompts that instruct large language models (LLMs) to interpret financial texts and generate semantic outputs. We construct customized prompts for three categories of financial inputs: news headlines, financial statement summaries, and macroeconomic event descriptions. Each prompt corresponds to a distinct semantic task. For example, prompts guide the

model to classify sentiment from news articles, to assess corporate outlook based on financial reports, and to evaluate economic tone from macro-level narratives. These prompts enable the model to perform semantic labeling consistently across different financial domains without requiring any additional training or fine-tuning.

(3) **LLM-Derived Signal Structuring Module:** This module transforms the raw semantic outputs produced by the LLM into structured numerical signals suitable for downstream modeling. Text-based labels such as "positive", "neutral", and "negative" are mapped to corresponding numeric values such as +1, 0, and -1. From these values, we derive additional features including daily sentiment scores, short-term trend indicators, and content volume statistics. These structured semantic signals provide a consistent and machine-readable representation of financial narratives, forming the foundation for subsequent stock return prediction and portfolio strategy modeling.

(4) **Stock Movement Prediction Module:** In this module, the extracted semantic signals are integrated with structured inputs, such as technical indicators and historical stock data. These combined features are fed into machine learning and deep learning models to predict expected returns. The models are evaluated using regression metrics including Mean Squared Error (MSE) and the coefficient of determination (R^2). In addition, signal-based backtesting and factor analysis are conducted to assess the real-world effectiveness and interpretability of the semantic and technical signals in guiding stock selection decisions.

3.1 Data Construction Module

The goal of this module is to obtain and prepare all financial data required for downstream analysis. The process consists of three parts: data collection, data cleaning, and data-to-text transformation. We use the EODHD API to retrieve structured data, which is then cleaned

and converted into natural language format, enabling prompt-based inference in downstream modules using large language models (LLMs).

3.1.1 Data Collection

This study aims to develop a stock selection framework that integrates semantic understanding from large language models (LLMs) guided by prompt engineering. The goal is to extract predictive signals from financial texts through zero-shot inference. We collect both structured and unstructured financial data via the EODHD API, a widely used platform for accessing historical and real-time market information. The constructed dataset consists of four major components:

1. Stock price data: Historical stock prices of S&P 100 companies, including open, high, low, close, volume, and adjusted close, are collected via the EODHD API. In addition, we retrieve technical indicators such as moving averages, RSI, and MACD to capture price trends and momentum features.

2. Financial news: We gather time-stamped news articles associated with S&P 100 companies via the EODHD News API. Each article includes metadata such as the publication time, title, full content, and other related information. These articles serve as the primary unstructured text input for sentiment analysis and semantic classification in downstream tasks.

3. Financial statements: We collect quarterly financial statements of S&P 100 companies through the EODHD Fundamentals API. These include key items from income statements, balance sheets, and cash flow statements, such as revenue, net income, total assets, shareholders' equity, and free cash flow.

4. Macroeconomic data: We collect macroeconomic events and indicators, such

as interest rate decisions, inflation reports, and GDP updates, via the EODHD Macro Event API. These data are used to represent the broader economic environment and are incorporated into the model to account for macro-level influences on stock price movements.

3.1.2 Data Cleaning

To ensure data quality and minimize noise in textual inputs, we apply a two-level cleaning process for financial news data. At the data level, we remove duplicate records, invalid dates, and entries with missing or insufficient textual content. At the content level, we normalize and tokenize the text by removing special characters, URLs, non-ASCII symbols, and stop words. Additionally, we implement a semantic deduplication method using TF-IDF vectorization and cosine similarity to eliminate nearly duplicate news articles. These procedures help improve the consistency and clarity of the data used for downstream modeling.

3.1.3 Data-to-Text Transformation

To enable LLMs to process structured financial data, we perform a data-to-text transformation procedure. This step focuses on converting numerical values from financial statements and macroeconomic indicators into natural language descriptions. For example, key metrics such as revenue growth, changes in net income, or interest rate adjustments are expressed using template-based sentences to convey meaningful semantics. The resulting text serves as input to large language models for semantic inference tasks, guided by task-specific prompts.

Earnings Report Template

Note: Company names and actual reporting dates are masked.

The following is a summary of a company ' s financial performance for this quarter.

Category	Metrics
Operational Performance	Revenue: [amount] USD; Gross Profit: [amount] USD; Gross Margin: [percentage] (QoQ: [change], YoY: [change])
Profitability	Net Income: [amount] USD; Operating Income: [amount] USD; EPS: [amount] USD; ROE: [percentage] (QoQ: [change], YoY: [change])
Financial Structure	Total Assets: [amount] USD; Total Liabilities: [amount] USD; Equity: [amount] USD; D/E Ratio: [value]
Cash Flow	Free Cash Flow: [amount] USD (YoY: [change]); Operating Cash Flow: [amount] USD; CapEx: [amount] USD

Table 1: Quarterly Financial Summary (Anonymized Example)

The above table is for illustrative purposes only. In practice, the input to the large language model is a natural language paragraph conveying the same financial indicators, organized in a structured textual format to facilitate semantic inference.

Macroeconomic Event Template

Note: Actual publication dates have been masked

To account for differences in data availability, we define two versions of the macroeconomic event template:

(1) If forecast data is available: The [Indicator Name] came in at [Actual Value], versus a market expectation of [Forecast Value] and a previous reading of [Previous Value], reflecting a [Delta Value] change.

(2) If not: The [Indicator Name] came in at [Actual Value], with a previous reading of [Previous Value], reflecting a [Delta Value] change.

This flexible design maintains a consistent narrative structure while adapting to the level of information available. It allows LLMs to perform robust semantic inference even in cases of incomplete market data.

3.2 Prompt Engineering Module

This module focuses on the design of task-specific prompts to guide large language models (LLMs) in performing semantic inference on financial texts. We develop distinct prompt formats for three types of financial content: company news, financial statement summaries, and macroeconomic event narratives. Each prompt is tailored to elicit specific semantic judgments from the model, such as sentiment, corporate outlook, or macroeconomic interpretation. These prompts enable zero-shot classification and reasoning without the need for model fine-tuning.

3.2.1 News Prompt Engineering

For company-specific news, we design prompts that ask the LLM to classify sentiment based on the given headline. The model is expected to return one of three sentiment categories: *positive*, *neutral*, or *negative*. These prompts enable the model to interpret real-world financial news and infer market tone in a zero-shot manner.

Prompt (system message):

You are a professional financial market analyst. Your task is to classify the sentiment of the following news headline.

The sentiment should be one of: *positive*, *negative*, or *neutral*.

Only output the final answer in this format:

Final answer: [positive / negative / neutral]

Return only the sentiment of the news headline.

Prompt (user message): News headline: "[headline]"

3.2.2 Financial Report Prompt Engineering

To enhance interpretability and align the output with the reasoning process of human analysts, we adopt a chain-of-thought prompting strategy in the financial report task. Instead of directly requesting a label, the prompt guides the LLM to generate step-by-step reasoning based on key financial indicators, recent trends (QoQ, YoY), and inferred business momentum. The final output consists of a structured explanation followed by an outlook classification: *positive*, *neutral*, or *negative*.

Prompt (system message):

You are a professional equity analyst. Please analyze the following company's financial summary and assess its potential future outlook based solely on the data provided.

Your response must follow this format:

Step-by-step reasoning:

1. Interpreting key metrics (e.g., revenue, profit, margins, cash flow)
2. Identifying recent trends and changes (QoQ, YoY)
3. Inferring potential business momentum, risks, or signals of inflection

Final assessment: [positive / neutral / negative]

Prompt (user message): Financial Summary: "[summary]"

3.2.3 Macro Prompt Engineering

To capture the market implications of macroeconomic events, we construct task-specific prompts that guide large language models (LLMs) in interpreting macro-level financial developments. Each input consists of multiple headlines or article snippets published on the same day, all related to a specific event such as central bank policy announcements, inflation updates, or labor market reports.

To enhance interpretability and align the reasoning process with that of professional analysts, we adopt a chain-of-thought prompting strategy. Instead of requesting a sentiment label directly, the prompt guides the LLM to reason step by step through the macroeconomic context and its potential market implications. The final output consists of a structured explanation followed by a sentiment classification: *positive*, *neutral*, or *negative*.

Prompt (system message):

You are a professional financial market analyst. Please analyze the following macroeconomic data and assess its likely impact on the financial markets.

Your response must follow this format:

Step-by-step reasoning:

1. Interpreting the macroeconomic context (e.g., interest rate decisions, inflation reports, employment data)
2. Comparing with prior values or market expectations
3. Inferring the likely impact on financial indicators (e.g., growth, rates, inflation outlook)
4. Assessing the potential short-term market reaction (e.g., bullish, bearish, neutral)

Final assessment: [positive / neutral / negative]

Prompt (user message): Macro Summary: "[macro]"

3.3 LLM-Derived Signal Structuring Module

This module constructs structured semantic features from the outputs of large language models (LLMs) applied to three types of financial texts: company news, financial report summaries, and macroeconomic event narratives. Rather than applying model fine-tuning, we utilize task-specific prompts to guide the LLM in performing semantic classification for each text type. The resulting textual outputs are mapped to numerical values and further processed to generate time-series features. These structured signals, aligned at the daily or quarterly level, serve as model-ready inputs for subsequent prediction and portfolio analysis.

3.3.1 News Sentiment Analysis

For company-specific news, the language model assigns each article a sentiment label: positive, neutral, or negative. These labels are mapped to numerical values (+1, 0, -1).

The daily sentiment score is calculated as the average of all sentiment values on that trading day. This score serves as a core semantic signal.

Feature	News Sentiment Score (S_t)
Definition	The average sentiment label of all company-specific news on day t , where labels are mapped as: +1 (positive), 0 (neutral), -1 (negative).
Usage	Core indicator representing the daily tone of news related to a specific stock.
Formula	$S_t = \frac{1}{N_t} \sum_{i=1}^{N_t} y_i$, where $y_i \in \{-1, 0, +1\}$

Table 2: News Sentiment Score (S_t)

To better capture temporal sentiment dynamics, we construct three auxiliary features: the moving average of sentiment scores (reflecting short-term trends), the daily change in sentiment (capturing momentum or reversals), and daily news volume, which serves as a proxy for information intensity and market attention.

Feature	News Sentiment Moving Average (MA_t^S)
Definition	The average of sentiment scores over the past n trading days.
Usage	Smooths daily fluctuations to capture short-term sentiment trends.
Formula	$MA_t^S = \frac{1}{n} \sum_{i=0}^{n-1} S_{t-i}$

Table 3: News Sentiment Moving Average (MA_t^S)

Feature	News Sentiment Change (ΔS_t)
Definition	Day-to-day difference in sentiment score.
Usage	Captures momentum or reversals in market tone.
Formula	$\Delta S_t = S_t - S_{t-1}$

Table 4: News Sentiment Change (ΔS_t)

Feature	Daily News Volume (N_t)
Definition	The total number of news articles related to a given stock or market on day t .
Usage	Serves as a proxy for information intensity or event-driven attention. Higher values may indicate abnormal news coverage, market focus, or external shocks.
Formula	$N_t =$ Total number of news articles on day t

Table 5: Daily News Volume (N_t)

3.3.2 Financial Report Commentary Analysis

Each quarterly earnings report is converted into multiple natural language statements summarizing changes in key financial indicators, such as revenue, net income, ROE, and operating cash flow. These statements are classified by the language model via prompt-based inference as positive, neutral, or negative, and mapped to numerical values of +1, 0, or -1. The average of these scores yields the Earnings Outlook Polarity Score (EOPS), which quantifies the overall semantic tone of the company ’ s financial disclosure.

Feature	Earnings Outlook Polarity Score (E_q)
Definition	Average sentiment score across all model-classified financial statement sentences.
Usage	Quantifies the overall semantic tone of a company ’ s earnings report.
Formula	$E_q = \frac{1}{n} \sum_{i=1}^n y_i$, where $y_i \in \{1, 0, -1\}$

Table 6: Earnings Outlook Polarity Score (EOPS)

To better track semantic shifts over time, we derive an auxiliary feature: the quarter-over-quarter change in EOPS, which captures how the company ’ s financial tone evolves across reporting

periods. This highlights directional sentiment changes that may influence market perception.

Feature	Quarter-over-Quarter Change (ΔE_q)
Definition	Difference in EOPS between the current and previous quarters.
Usage	Tracks semantic sentiment shifts across reporting periods.
Formula	$\Delta E_q = E_q - E_{q-1}$

Table 7: Quarter-over-Quarter Change in EOPS

3.3.3 Macroeconomic Event Analysis

For macroeconomic-related texts such as Federal Reserve decisions, GDP reports, or inflation announcements, each event description is treated as an independent input and is individually processed by a prompt-based language model for sentiment classification. The model assigns a sentiment label of positive, neutral, or negative, which is then mapped to a numerical value of +1, 0, or -1. The average of all sentiment scores for macroeconomic events on a given day represents the overall semantic tone of macroeconomic information for that day.

Feature	Macroeconomic Polarity Score (P_t^M)
Definition	The average sentiment score assigned to all macroeconomic event texts on day t , where each event is independently evaluated by the LLM and mapped to a polarity value.
Usage	Represents the overall semantic tone of macroeconomic information for a given day, aggregated from multiple event-level classifications.
Formula	$P_t^M = \frac{1}{M_t} \sum_{i=1}^{M_t} p_i$, where $p_i \in \{-1, 0, +1\}$

Table 8: Macroeconomic Polarity Score (P_t^M)

To further capture the temporal dynamics and informational intensity of macroeconomic sentiment, we extract three auxiliary features based on the daily polarity score: (1) the day-to-day change in score, which reflects short-term shifts in economic outlook; (2) the

moving average of polarity scores over a fixed window, which smooths fluctuations and highlights recent trends; and (3) the macro event volume, which serves as a proxy for macro-level information density and event-driven market attention.

Feature	Macro Sentiment Change (ΔM_t)
Definition	Difference between today's and yesterday's macroeconomic polarity score.
Usage	Reflects day-to-day variation in macroeconomic sentiment.
Formula	$\Delta M_t = M_t - M_{t-1}$

Table 9: Macro Sentiment Change (ΔM_t)

Feature	Macro Moving Average (MA_t^M)
Definition	Moving average of macroeconomic polarity scores over the past n trading days.
Usage	Reveals short-term macro sentiment trends.
Formula	$MA_t^M = \frac{1}{n} \sum_{i=0}^{n-1} M_{t-i}$

Table 10: Macro Moving Average (MA_t^M)

Feature	Macro Event Volume (M_t)
Definition	The total number of macroeconomic event descriptions processed on day t .
Usage	Proxy for macro-level information density and event-driven market influence. Higher volumes may reflect increased policy activity or economic uncertainty.
Formula	$M_t = \text{Number of macroeconomic event texts on day } t$

Table 11: Macro Event Volume (M_t)

3.4 Stock Movement Prediction Module

This module integrates semantic features inferred from prompt-based large language models (LLMs), along with technical indicators and historical price data, to perform stock price movement prediction. By combining unstructured textual signals with structured financial data, we train machine learning and deep learning models to capture complex behavioral patterns in the market. The objective is to assess whether this multi-source feature integration improves predictive accuracy and enhances model interpretability in real-world forecasting scenarios.

3.4.1 Stock Movement Prediction

We train prediction models using a combined feature set composed of three major sources. First, **semantic features** are inferred from prompt-based large language models, including news sentiment scores and their moving averages, earnings outlook polarity scores and quarter-over-quarter changes, as well as macroeconomic polarity scores and daily variations. Second, we include **technical indicators** that capture price trends and momentum, such as moving averages, RSI, MACD, and Bollinger Bands. Finally, **price-based features** are incorporated, including daily open, high, low, and close (OHLC) prices.

These features are aligned at the daily level and used as inputs to train a variety of regression models, including XGBoost, LSTM, and multilayer perceptron (MLP) networks. The prediction target is the logarithmic return over a specified future horizon. Prediction performance is evaluated using Mean Squared Error (MSE).

Feature	Next-Day Log Return (r_{t+1})
Definition	The logarithmic return between the closing prices of day t and $t + 1$, defined as:
	$r_{t+1} = \log \left(\frac{P_{t+1}}{P_t} \right)$
	where P_t is the closing price on day t .
Usage	Serves as the target variable for return prediction models. Log returns are preferred due to their additive property across time and their statistical convenience for modeling financial time series.

Table 12: Definition of Next-Day Log Return (r_{t+1})

3.4.2 LLM-Derived Signal Backtesting

To evaluate the real-world applicability of the extracted signals, we conduct a backtesting procedure based on a rule-based long-short strategy. In this stage, we do not predict future prices directly. Instead, we use the semantic and technical signals to generate trading decisions based on fixed thresholds.

At the end of each trading week, all S&P 100 stocks are evaluated against preset semantic conditions, such as sentiment score thresholds from news or earnings analysis. Stocks that meet bullish conditions are added to the long candidate pool, while those meeting bearish conditions are added to the short pool. From each pool, a fixed number of stocks (e.g., top 10) are selected to construct a dollar-neutral portfolio.

The portfolio is rebalanced weekly based on updated semantic signals. Macroeconomic

sentiment is used to adjust overall exposure, reducing position size during negative outlooks and increasing leverage under favorable conditions. Strategy performance is evaluated using standard portfolio-level metrics, including cumulative return, Sharpe ratio, volatility, and maximum drawdown.

Chapter 4. Experiments

This chapter presents the implementation of the research framework introduced in Chapter 3, focusing on evaluating the effectiveness of semantic features through predictive accuracy and investment performance. Semantic features are extracted from financial texts, including news sentiment obtained via sentiment analysis, and earnings outlook and macroeconomic outlook inferred through prompt-based reasoning.

The experiments are conducted on the full S&P 100 stock universe. These semantic signals are combined with technical indicators and price-based features to form a multi-source feature set.

The experimental evaluation consists of two parts. First, we assess whether the integrated feature set improves the accuracy of stock return prediction using supervised learning models such as XGBoost, LSTM, and multilayer perceptrons (MLPs). Second, we evaluate the investment value of the extracted signals through a backtesting simulation, where stocks are directly ranked and selected based on semantic signals without relying on machine learning. Long-short portfolios are formed accordingly, and strategy performance is evaluated using portfolio-level metrics such as cumulative return, Sharpe ratio, and maximum drawdown.

4.1 Data Construction Module

4.1.1 Data Collection

We constructed a multi-source dataset covering all S&P 100 stocks from January 2020 to March 2025. The dataset includes structured and unstructured financial data collected through APIs and processed for downstream analysis. A summary of the collected datasets is shown in Table.

Dataset Type	Time Range	Description
News Headlines	2020-01 to 2025-03	A total of 639,902 headlines across 100 companies, used for sentiment classification.
Earnings Commentary	2020 Q1 to 2025 Q1	21 reports per company, totaling 2,100 quarterly summaries transformed into natural language for outlook classification.
Macroeconomic Indicators	2020 Q1 to 2025 Q1	19,968 records across 413 distinct indicators, converted into narrative event texts for semantic inference.
OHLCV Stock Data	2020-01 to 2025-03	Approximately 130,000 daily trading records, including open, high, low, close, and volume.
Technical Indicators	2020-01 to 2025-03	Includes 5MA, RSI, MACD, and BB% indicators.

Table 13: Overview of Collected Datasets

4.1.2 Data Cleaning

We applied basic cleaning procedures to the news dataset, including the removal of duplicates, correction of invalid or missing entries, and filtering of irrelevant or noisy content. For macroeconomic data, we selected 60 key indicators from the original set to focus on the most relevant economic signals.

Dataset Type	Original Records	Cleaned Records	Coverage
News Headlines	639,902	528,556	100 stocks
Macroeconomic Indicators	19,968	8,341	60+ indicators

Table 14: Summary of Data Cleaning Results

The cleaned data is of sufficient quality to support subsequent modeling and analysis.

4.1.3 Data-to-Text Transformation

Among the various data sources used in this study, only earnings reports) and macroeconomic indicators required transformation from structured tables into natural language format. This process was necessary to allow LLMs to interpret structured numerical values using their language understanding and instruction-following capabilities.

Earnings Report Transformation

Each earnings record contains structured fields such as EPS (actual), EPS (expected), and revenue. These values were mapped into templated English sentences and served as input prompts for the LLM to infer overall sentiment or performance tone. If previous quarter data was available, the text also includes a brief comparison to highlight quarter-over-quarter changes. In total, 2,100 earnings summaries were generated.

Example:

The company reported quarterly revenue of \$5.86 billion, with a gross profit of \$3.26 billion and a gross margin of 55.72%, reflecting a QoQ increase of 1.42% and a YoY growth of 1.19%. Net income was \$385 million, operating income reached \$714 million, and EPS stood at \$0.09. ROE was 1.58%, with net income declining 54.55% QoQ and 18.26% YoY. Total assets were \$81.01 billion, liabilities \$56.61 billion, and equity \$24.40 billion, resulting in a debt-to-equity ratio of 2.32. Free cash flow was \$2.21 billion (YoY +8.90%), while operating cash flow was \$2.32 billion and CapEx amounted to \$108 million.

Macroeconomic Event Text Transformation

For each macroeconomic indicator release (e.g., CPI, unemployment rate, ISM PMI, GDP), structured numerical data was transformed into natural language sentences to enhance the interpretability and reasoning capabilities of large language models. When both forecasted values and prior period data were available, the system applied a conditional template to generate comparative descriptions. If any key value was missing, the sentence was automatically simplified to maintain fluency. In total, 8,341 macroeconomic event sentences were generated from over 60 selected indicators.

Example (GDP): The U.S. GDP Price Index (QoQ Annualized, Advance Estimate) was 1.5%, below the market expectation of 1.8% and the previous value of 1.7%, reflecting a -0.2% change.

4.2 Prompt Engineering Module

In this study, we utilize the Llama-3.3-70B-Versatile language model developed by Meta as the primary engine for prompt-based semantic classification tasks. Based on the Llama 3.3 architecture, this model comprises 70 billion parameters and incorporates an enhanced transformer design. It employs Grouped-Query Attention to boost inference efficiency and scalability. Additionally, the model has been fine-tuned through supervised learning) and further refined using reinforcement learning with human feedback, enabling better alignment with human preferences in terms of usefulness, coherence, and safety.

We selected this model due to its strong performance across a variety of natural language processing benchmarks, including 86.0% accuracy on MMLU (Massive Multitask Language Understanding), 88.4% pass@1 on HumanEval (code generation), 77.0% SymPy intersection score on MATH (mathematical reasoning), and 91.1% exact match on MGSM (Multilingual Grade School Math). These findings highlight the model ’ s overall strength in reasoning and language comprehension, indicating its suitability for zero-shot financial document analysis tasks.

4.2.1 News Prompt Engineering

To extract sentiment from financial news, we applied a zero-shot prompt using the headline alone as input. The model was instructed to classify the sentiment into one of three categories: positive, neutral, or negative. Following the design described in Chapter 3, we implemented a role-based prompt structure, consisting of a system message that defines the task and a user message that provides the headline. For each headline, the model returned a single sentiment label. These outputs were collected as raw semantic

annotations for downstream use.

To assess how prompt structure impacts output formatting, we compared a format-constrained prompt (as introduced in Chapter 3) with a commonly used unconstrained prompt in the context of financial news sentiment classification.

Aspect	Unconstrained Prompt	Format-Constrained Prompt
Prompt Example	<i>Classify the sentiment of this headline: "Foxconn 's profit down 24%"</i>	As designed in Section 3.2.1
Model Output	<i>Negative. Because the headline...</i>	<i>Final answer: negative</i>
Format Consistency	Varies in length and style	Consistent and structured
Usability	Needs post-processing	Ready for direct use

Table 15: Comparison of prompt styles for news sentiment

This comparison highlights the robustness of the format-constrained prompt. By enforcing a consistent output style, it reduces ambiguity and enhances scalability for automated sentiment analysis of financial news headlines.

The sentiment classification prompt was applied to 528,556 financial news headlines, with the model successfully producing a sentiment label for each. The outputs were consistent and well-structured, confirming the method 's scalability for real-world, high-volume financial data.

4.2.2 Financial Report Prompt Engineering

In this section, we use a prompt tailored for earnings reports to extract forward-looking sentiment. Each input is a natural language summary of key financial metrics such as revenue, profit, and cash flow. To enhance interpretability, we guide the model to reason step-by-step (Chain of Thought) before selecting a sentiment label: positive, neutral, or negative.

To evaluate how prompt structure influences reasoning quality and output consistency in forward-looking financial sentiment classification, we compared two prompt versions: one without reasoning guidance and one with a structured Chain-of-Thought (CoT) approach.

Aspect	Without CoT	With Cot
Prompt Design	Vague, no structure or steps	Step-by-step reasoning required
Output Format	Long, unstructured text	Structured and concise
Label Clarity	Often missing or unclear	Clearly stated (e.g., <i>neutral</i>)
Content Quality	Mixed signals, no conclusion	Logical reasoning with key highlights
Example Excerpt	““Overall,the company ’ s financial performance is ..””	Based on revenue, margin trends... Final assessment: neutral

Table 16: Comparison of prompt strategies for financial sentiment

This comparison demonstrates that adding structured reasoning not only improves the

clarity and consistency of the output, but also ensures it aligns with the task ' s objective and is ready for scalable automation.

We applied the prompt-based classification framework to 2,100 earnings report summaries. For every instance, the model successfully returned a sentiment label. In the Chain-of-Thought setting, the output included structured, step-by-step reasoning traces, demonstrating the model ' s ability to generate interpretable and consistent financial sentiment judgments across real-world earnings narratives.

4.2.3 Macro Prompt Engineering

This section outlines the application of prompt-based classification to macroeconomic event texts. Each input consists of a sentence describing an economic event, such as an interest rate decision, an inflation report, or a GDP release. The prompt instructs the language model to evaluate the potential impact on financial markets and return a sentiment label: positive, neutral, or negative.

To ensure consistency across inputs, all events are rewritten as concise natural language summaries. We further investigate whether prompting the model with step-by-step reasoning (Chain-of-Thought) helps produce outputs that are more interpretable and consistent.

As shown in the table above, prompts with structured reasoning significantly improve clarity, label consistency, and alignment with task requirements, making them better suited for automation and large-scale analysis.

In total, we applied this prompt-based classification framework to 8,341 macroeconomic event texts. For every instance, the model successfully returned a sentiment label, and in the Chain-of-Thought setting, we observed clear step-by-step reasoning traces. This confirms the method ' s scalability and consistency in real-world data applications.

Aspect	Without CoT	With CoT
Structure	Unstructured single paragraph	Step-by-step reasoning with clear format
Explanation Depth	Moderate; implicit reasoning, loosely organized	High; explicit and sequential reasoning steps
Label Clarity	Often ambiguous (e.g., <i>neutral to slightly negative</i>)	Clear and consistent (e.g., <i>Final assessment: negative</i>)
Automation Suitability	Low; requires manual parsing	High; label can be directly extracted
Market Reasoning Coverage	General discussion; lacks specificity	Covers inflation, interest rates, bonds, currencies

Table 17: Comparison of prompt strategies for macro sentiment

4.3 Financial Text Signal Structuring Module

4.3.1 News Sentiment Analysis

To extract semantic signals from financial news, we adopt a zero-shot prompting approach that classifies each news headline as positive, neutral, or negative. These sentiment labels are then converted into numerical values: +1 for positive, 0 for neutral, and -1 for negative.

Based on this, we compute four daily semantic features:

- (1) **News Sentiment Score**: the average sentiment of all headlines on a given day;
- (2) **Sentiment Moving Average**: the average sentiment over the past n days, capturing short-term trends;
- (3) **Sentiment Change**: the day-to-day difference in sentiment, indicating momentum or reversals;
- (4) **News Volume**: the number of company-related news articles on that day, serving as a proxy for information intensity.

After constructing the semantic features, we proceed with further analysis and validation.

Distribution of sentiment for 100 stocks between 2020 and 2025

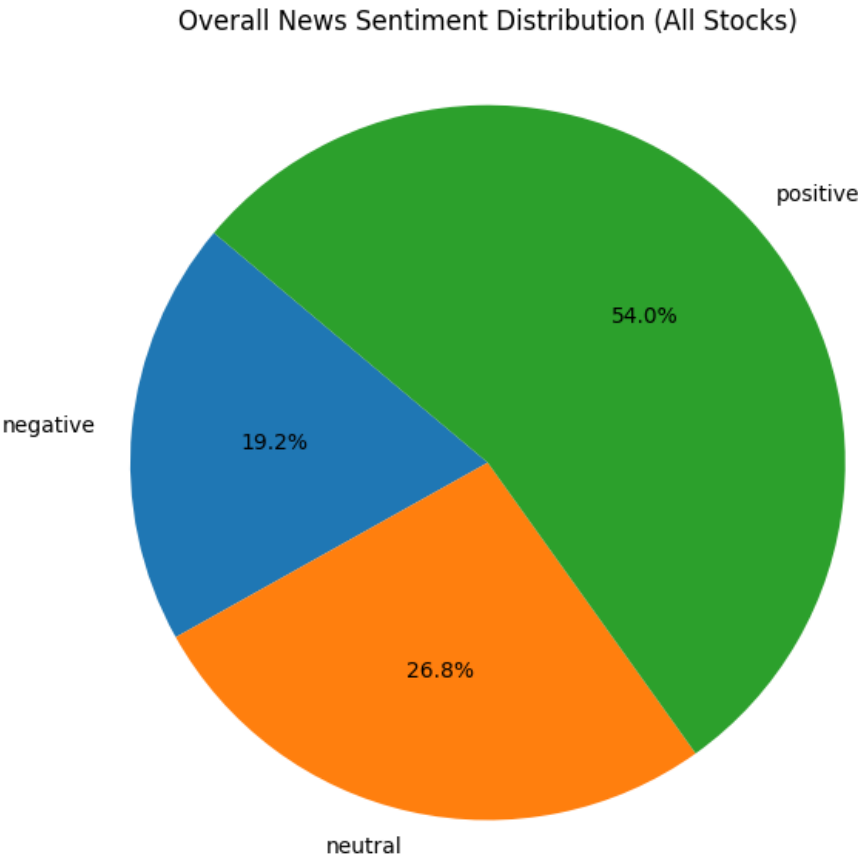


Figure 3: Distribution of sentiment for 100 stocks

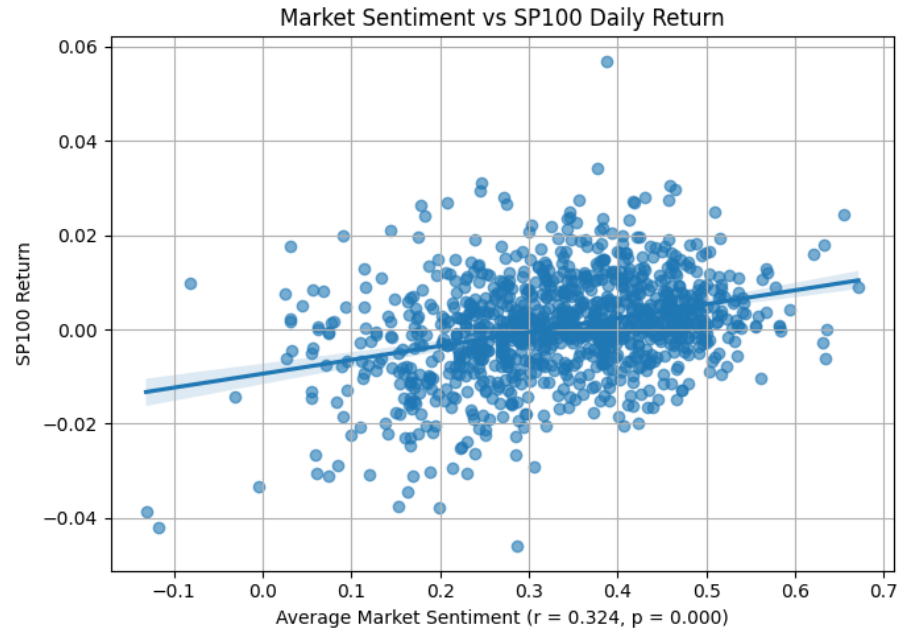


Figure 4: Market Sentiment vs. SP100 Return (News Volume 50)

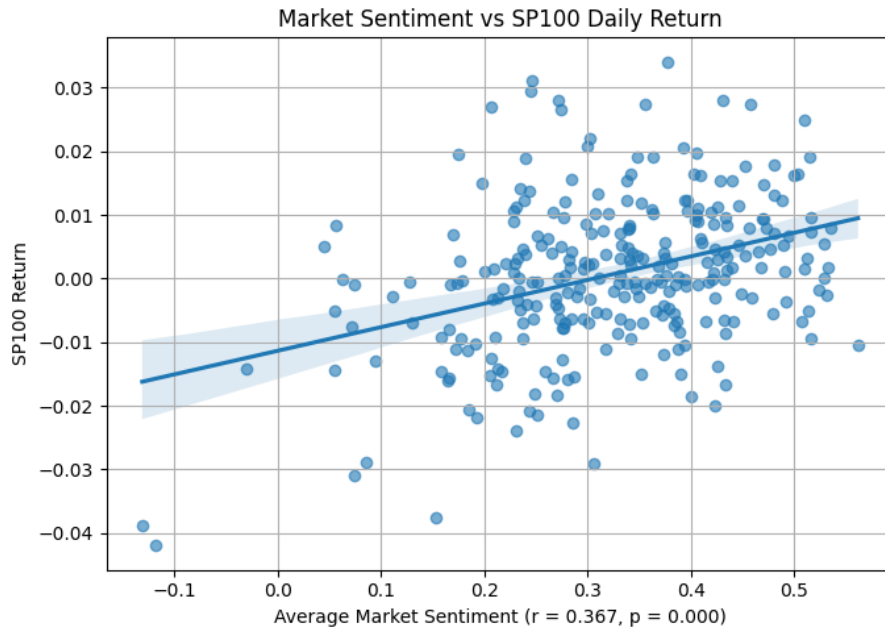


Figure 5: Market Sentiment vs. SP100 Return (News Volume 500)

To evaluate the relationship between news sentiment and price movement, we filtered trading days based on the minimum number of news articles. The correlation between average sentiment score and daily return increases as the news volume threshold rises:

News Count Threshold	Sentiment-Return Correlation
≥ 0	0.100
≥ 10	0.210
≥ 50	0.324
≥ 100	0.335
≥ 150	0.358
≥ 500	0.367

Table 18: Correlation between daily average news sentiment and stock returns under different news volume thresholds

These results suggest that sentiment signals extracted by LLMs are more predictive on

days with higher news volume, indicating stronger alignment between textual sentiment and market response in high-information environments.

4.3.2 Financial Report Commentary Analysis

To extract semantic signals from company financial disclosures, we adopt a zero-shot prompting approach that classifies each sentence in an earnings report as positive, neutral, or negative. These sentiment labels are then converted into numerical values: +1 for positive, 0 for neutral, and -1 for negative.

Based on this, we compute two quarterly semantic features:

(1) **Earnings Outlook Polarity Score:** the average sentiment across all classified sentences within a company's financial statement for a given quarter;

(2) **Quarter-over-Quarter Change:** the difference in polarity scores between the current and previous quarters, capturing shifts in financial tone over time.

These features quantify not only the static tone of a given earnings report but also dynamic changes in corporate sentiment that may influence market expectations.

With the semantic features in place, we then conduct detailed analysis to evaluate their effectiveness.

Sentiment Distribution (Manually Set Proportions)

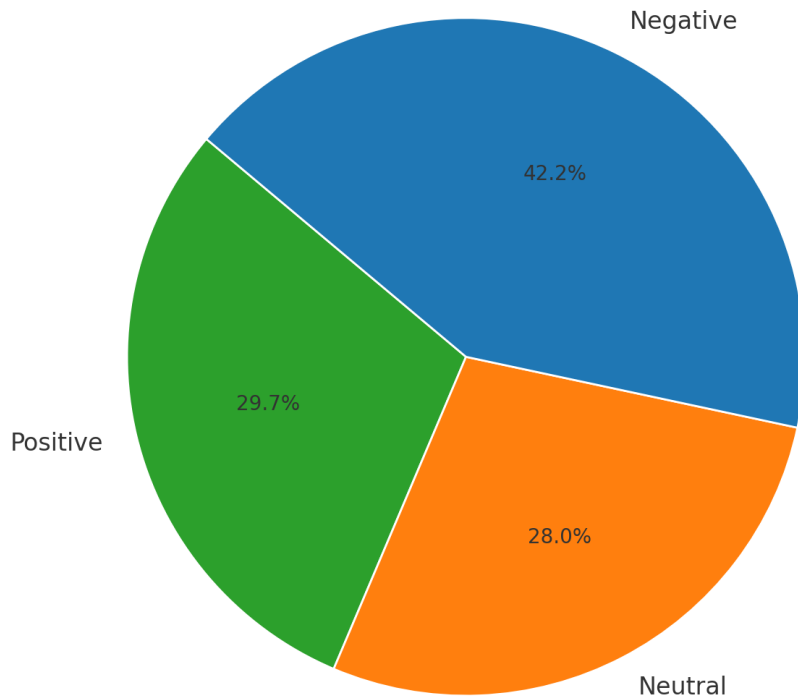


Figure 6: Distribution of financial outlook for 100 stocks

In the raw sentiment dataset, negative tone accounted for 42.2%, positive for 29.7%, and neutral for 28.0%.

After filtering for valid events with matched stock price data and excluding neutral labels, the final analysis sample contained 1,381 events, with 59.0% labeled as positive and 41.0% as negative.

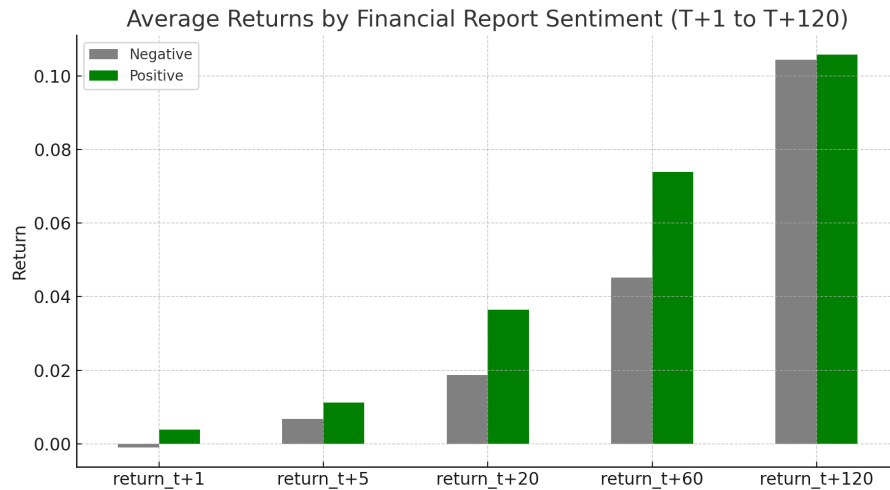


Figure 7: Average Returns by Financial Outlook

Figure Average future stock returns following financial reports with positive vs. negative sentiment, evaluated at horizons of 1, 5, 20, 60, and 120 trading days.

Sentiment was derived from text-based analysis of earnings reports.

The figure shows that positive sentiment is associated with significantly higher short- to mid-term returns, especially at T+20 and T+60. The effect fades out by T+120.

1. Average Return (Positive vs. Negative)

Forecast Horizon	Negative Outlook	Positive Outlook
T+1	-0.10%	+0.38%
T+5	+0.68%	+1.13%
T+20	+1.87%	+3.65%
T+60	+4.52%	+7.39%
T+120	+10.4%	+10.6%

Table 19: Average returns following negative vs. positive Outlook across forecast horizons

2. Win Rate (Positive vs. Negative)

Forecast Horizon	Negative Outlook	Positive Outlook
T+1	39.4%	48.0%
T+5	58.9%	64.6%
T+20	57.1%	63.5%
T+60	58.6%	71.3%
T+120	65.5%	68.6%

Table 20: Upward movement probability under negative vs. positive Outlook across forecast horizons

3. t-test (Difference in Returns: Positive vs. Negative)

Forecast Horizon	t-value	p-value	Interpretation
T+1	4.54	0.000006	Significant
T+5	2.11	0.035	Significant
T+20	3.40	0.0007	Significant
T+60	3.05	0.0024	Significant
T+120	0.10	0.9203	Not significant

Table 21: t-test results for average return differences across forecast horizons

To assess the predictive power of earnings report sentiment, we analyzed 1,381 financial report events across 100 companies.

We computed stock returns from the report date to various horizons: T+1, T+5, T+20, T+60, and T+120.

The results indicate that **positive sentiment is associated with significantly higher**

average returns than negative sentiment at all horizons up to T+60.

For instance, at T+20, the average return following positive sentiment was **+3.65%**, compared to **+1.87%** for negative sentiment ($t = 3.40, p = 0.0007$).

At T+60, the gap remained significant ($t = 3.05, p = 0.0024$).

However, the difference was no longer statistically significant by T+120, suggesting that the market gradually absorbs and prices in the information contained in financial report tone.

4.3.3 Macroeconomic Event Analysis

To extract semantic signals from macroeconomic text, we adopt a zero-shot prompting approach that classifies each event description as positive, neutral, or negative. These sentiment labels are then converted into numerical values: +1 for positive, 0 for neutral, and -1 for negative.

Based on this, we compute four daily semantic features:

(1) **Macroeconomic Polarity Score**: the average sentiment score of all macroeconomic event texts on a given day;

(2) **Macro Outlook Moving Average**: the average polarity over the past n trading days, capturing short-term sentiment trends;

(3) **Macro Outlook Change**: the day-to-day difference in polarity, indicating shifts in economic outlook;

(4) **Macro Event Volume**: the number of macroeconomic texts processed on that day, serving as a proxy for information intensity.

After constructing the semantic features, we move on to the next stage of analysis.

Distribution of Daily Macro Sentiment (Negative / Neutral / Positive)

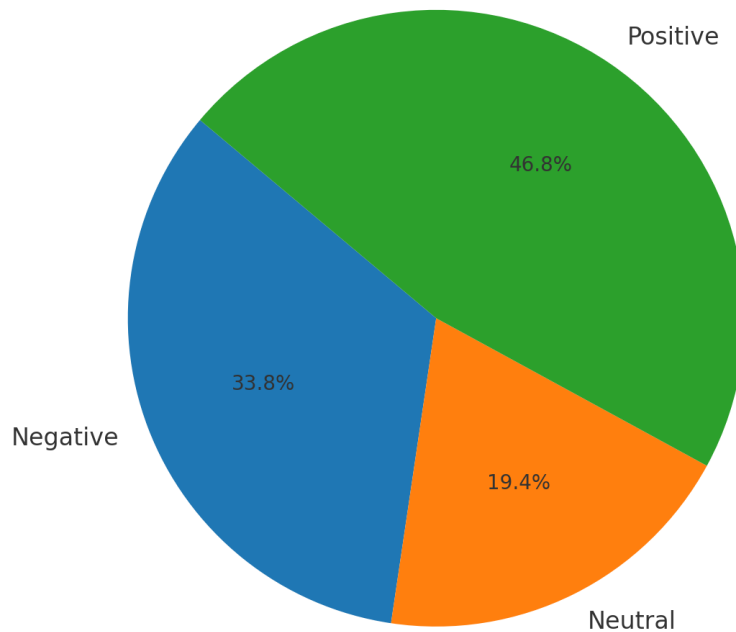


Figure 8: Distribution of Macro outlook for five yaers

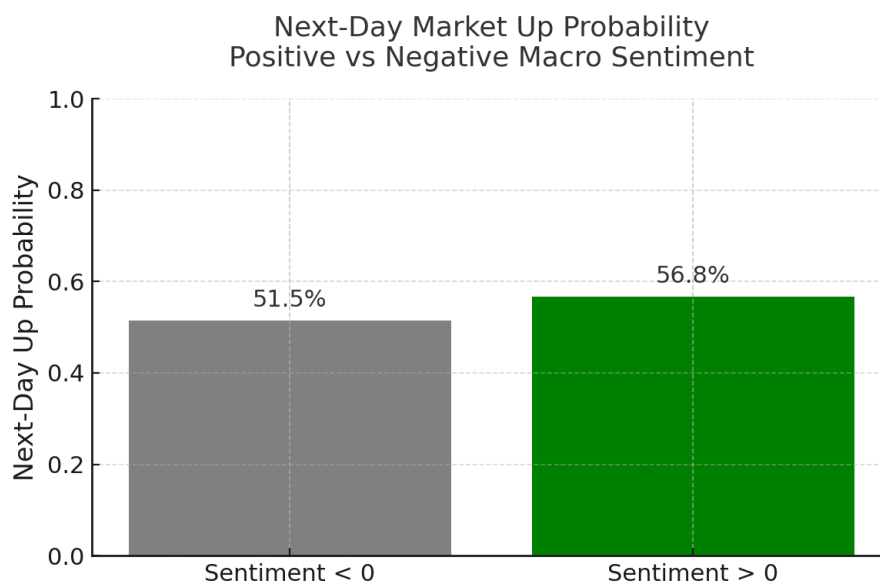


Figure 9: Next-Day Market Up Probability

To evaluate the predictive value of clearly positive or negative macroeconomic sentiment, we excluded neutral sentiment days and compared the next-day market up probability

between positive and negative sentiment groups.

The results show that days with positive sentiment were followed by a **higher up probability (56.8%)** compared to negative sentiment days (52.6%).

Statistical tests reveal a **marginally significant difference**: the t-test yields $p = 0.092$, and the chi-squared test gives $p = 0.104$, suggesting that the relationship holds at the **90% confidence level**, but not at the conventional 95% level.

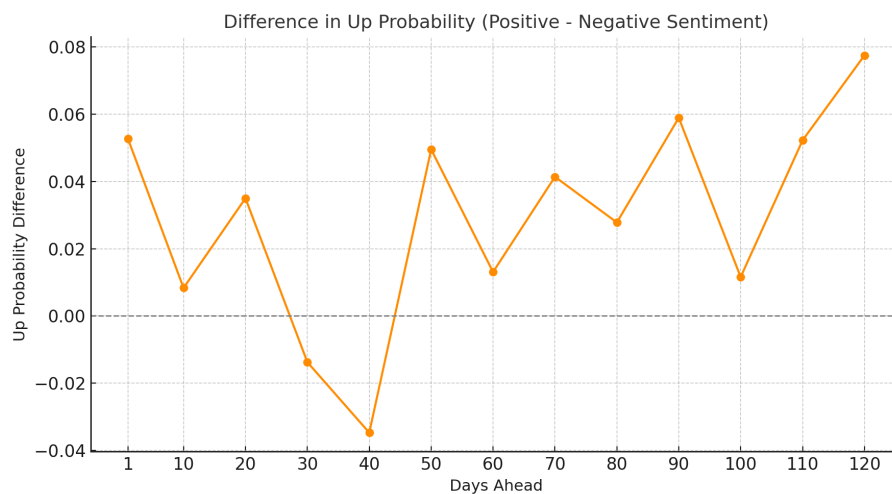


Figure 10: Difference in Up Probability

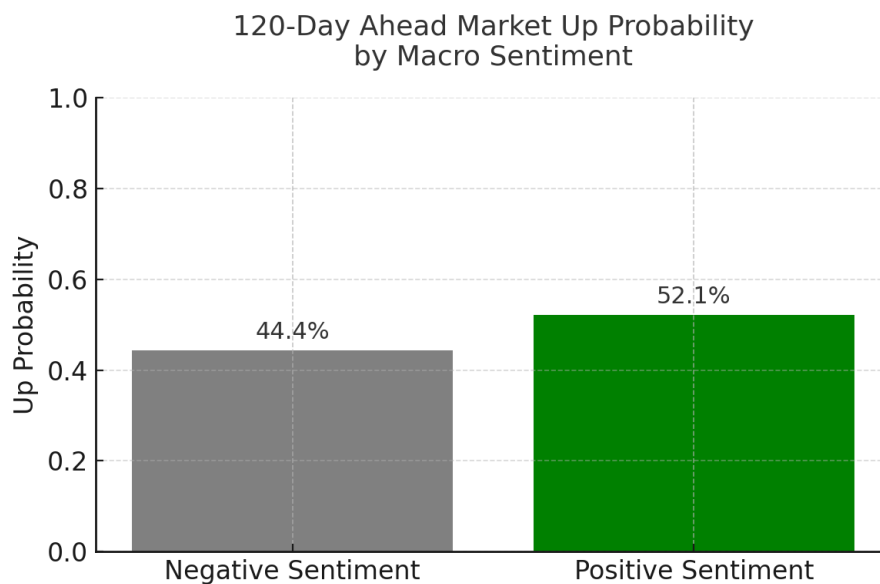


Figure 11: 120 Days Market Up Probability

Among all tested forecast horizons, the **120-day return** exhibited the most significant

divergence in performance between days with positive and negative macroeconomic sentiment.

Specifically, **437 negative sentiment days** had an average up probability of **44.4%**, while **606 positive sentiment days** reached **52.1%**.

This difference is statistically significant ($t = 2.48$, $p = 0.013$), suggesting that an optimistic macroeconomic tone may carry longer-term bullish implications.

4.4 Stock Movement Prediction Module

4.4.1 Stock Movement Prediction

This section outlines the implementation details of the stock return prediction task using semantic features. Three supervised learning models are employed: XGBoost, LSTM, and multilayer perceptron (MLP). The input consists of price information, technical indicators, and semantic features generated by large language model inference.

The target stocks include Apple (AAPL), JPMorgan Chase (JPM), Tesla (TSLA), Netflix (NFLX) and Johnson & Johnson (JNJ). The data period spans from January 2020 to March 2025. The training set covers January 2020 to May 2023, while the testing set spans from June 2023 to March 2025.

Samples are constructed using a sliding window of 16 or 32 trading days. The prediction target is the next-day (horizon = 1) or 5-day average future log return (horizon = 5), defined as:

Feature	Next-Day Log Return (r_{t+1})
Definition	The logarithmic return between the closing prices of day t and $t + 1$, defined as:
	$r_{t+1} = \log \left(\frac{P_{t+1}}{P_t} \right)$
	where P_t is the closing price on day t .
Usage	Serves as the target variable for return prediction models. Log returns are preferred due to their additive property across time and their statistical convenience for modeling financial time series.

Table 22: Definition of Next-Day Log Return (r_{t+1})

Each input contains 20 features. Price features include the daily open, high, low, and close prices. Technical indicators include RSI, MACD, Bollinger Band percentage, and log-volume change. Semantic features include four variables derived from news sentiment (average score, delta, moving average, and article count), two from earnings tone (EOPS and its quarter-over-quarter change), and four from macroeconomic sentiment (daily polarity score, its change, moving average, and macro event count).

Models are trained and tested on aligned samples independently for each stock. LSTM models are run 20 times to mitigate randomness. Mean Squared Error (MSE) is employed as an evaluation metric to assess how different feature sets affect predictive performance.

4.4.2 LLM-Derived Signal Backtesting

This section describes the rule-based long-short strategy designed using LLM-inferred semantic signals. The strategy does not involve predictive models or regression algorithms. Instead, trading actions are triggered directly based on threshold conditions applied to semantic features.

The backtesting universe consists of the S&P 100 constituents, covering the period from January 2020 to March 2025. The strategy checks all eligible stocks weekly and selects trades based on predefined semantic criteria.

Stocks with semantic scores, such as average news sentiment or earnings outlook polarity (EOPS), below a lower threshold (e.g., -0.2) are included in the short candidate pool. Those with scores above an upper threshold (e.g., $+0.2$) are included in the long candidate pool. Each week, the top 10 stocks in each pool are selected for position entry. Both the long and short portfolios are equally weighted, resulting in a dollar-neutral overall portfolio.

Macroeconomic sentiment, such as the average polarity score (PM), is used to adjust exposure. If the macro score is negative, the strategy reduces total position size. If positive, it increases leverage proportionally.

Positions are held for one week and rebalanced at the end of each week using the latest semantic signals. Strategy performance is evaluated using cumulative return, annualized Sharpe ratio, and maximum drawdown. Performance is compared against a passive, equally weighted S&P 100 portfolio to assess the practical viability of the semantic-based investment approach.

Chapter 5. Results and Evaluation

5.1 Stock Movement Prediction

In this section, we present the results of our experiments conducted on five representative stocks: AAPL, TSLA, JPM, NFLX, and JNJ. For each stock, we carry out a comprehensive set of ablation studies utilizing different feature set combinations to assess their impact. These include price and technical indicators (price+tech), news sentiment, macroeconomic variables, and financial statement features, evaluated under different model types and forecasting horizons.

Three predictive models are used in this study: LSTM, XGBoost, and Random Forest. Each model is trained using two sliding window sizes (16 and 32) and evaluated over two prediction horizons (1 day and 5 days). Model performance is measured using mean squared error (MSE).

The historical data is divided based on time. The training period ranges from January 1, 2020 to May 31, 2023, while the testing period covers June 1, 2023 to March 31, 2025. This ensures that all predictions are made on future data that the models have not seen during training, which simulates a realistic forecasting scenario.

The results are presented to show how different types of features contribute to prediction performance across various models and stocks. Selected tables are included in the main text to highlight key findings. All results are included in the appendix for review.

The following results are generated using a **LSTM** model:

Scenario	Horizon=1 (win16)	Horizon=5 (win16)	Horizon=1 (win32)	Horizon=5 (win32)
price+tech	0.1581	0.9447	0.1587	0.9745
price+tech+news	0.1599	0.9528	0.1611	0.9407
price+tech+macro	0.1576	0.8999	0.1653	0.9072
price+tech+fin	0.1736	0.9541	0.1709	0.9539
all	0.1670	0.9517	0.1711	0.9368

Table 23: MSE (%) comparison for TSLA under different scenario, window sizes, and horizons

Scenario	Horizon=1 (win16)	Horizon=5 (win16)	Horizon=1 (win32)	Horizon=5 (win32)
price+tech	0.0632	0.1806	0.0409	0.2507
price+tech+news	0.0598	0.2355	0.0493	0.3163
price+tech+macro	0.0469	0.1707	0.0398	0.3002
price+tech+fin	0.0479	0.1755	0.0437	0.2507
all	0.0523	0.2482	0.0301	0.2422

Table 24: MSE (%) comparison for JPM under different scenario, window sizes, and horizons

Table 23 and Table 24 illustrate that the LSTM model benefits significantly from the inclusion of macroeconomic variables, particularly for TSLA and JPM under medium-term forecasting horizons. This supports the model ' s ability to capture longer-term patterns embedded in economic indicators.

The following results are generated using an LSTM model.

Scenario	Horizon=1 (win16)	Horizon=5 (win16)	Horizon=1 (win32)	Horizon=5 (win32)
price+tech	0.0090	0.0517	0.0098	0.0600
price+tech+news	0.0084	0.0516	0.0091	0.0498
price+tech+macro	0.0084	0.0540	0.0101	0.0552
price+tech+fin	0.0092	0.0455	0.0090	0.0646
all	0.0089	0.0496	0.0096	0.0531

Table 25: XGBoost MSE (%) comparison for AAPL under different scenario, window sizes, and horizons

The following results are generated using an Random Forest model.

Scenario	Horizon=1 (win16)	Horizon=5 (win16)	Horizon=1 (win32)	Horizon=5 (win32)
price+tech	0.0101	0.0504	0.0097	0.0528
price+tech+news	0.0103	0.0442	0.0094	0.0484
price+tech+macro	0.0096	0.0491	0.0094	0.0511
price+tech+fin	0.0108	0.0507	0.0099	0.0526
all	0.0102	0.0442	0.0097	0.0458

Table 26: Random Forest MSE(%) comparison for AAPL under different scenario, window sizes, and horizons

Scenario	Horizon=1 (win16)	Horizon=5 (win16)	Horizon=1 (win32)	Horizon=5 (win32)
price+tech	0.0206	0.0984	0.0188	0.0994
price+tech+news	0.0169	0.0971	0.0167	0.0931
price+tech+macro	0.0181	0.0952	0.0176	0.0900
price+tech+fin	0.0203	0.0977	0.0182	0.0977
all	0.0182	0.0936	0.0179	0.0938

Table 27: Random Forest MSE(%) comparison for NFLX under different scenario, window sizes, and horizons

Furthermore, Tables 25–26, and 27 show that both XGBoost and Random Forest achieve noticeable reductions in MSE when news sentiment features are included, particularly

for stocks sensitive to market sentiment, such as AAPL and NFLX. This confirms the predictive value of textual sentiment data, even for non-sequential models.

These findings demonstrate that the contribution of different feature types is not uniform across models and stocks. While deep learning models such as LSTM excel at capturing long-term macroeconomic trends, tree-based models like XGBoost and Random Forest are better suited to incorporating discrete, short-term signals such as market sentiment. This highlights the importance of model-feature alignment in financial forecasting and suggests that hybrid approaches or model ensembles may further improve predictive performance.

5.2 LLM-Derived Signal Backtesting

Based on the empirical findings in Chapters 4 and 5.1, we observe that different categories of features contribute to different predictive horizons. News sentiment proves especially effective in short-term forecasting, while macroeconomic and financial statement features are more relevant for long-term trends. In addition, technical indicators such as RSI, MACD, and moving averages serve as essential filters to refine entry and exit timing.

Building on these insights, we propose a hybrid stock selection strategy that integrates semantic and technical signals. Short-term positions are guided primarily by news sentiment. A stock is selected for entry if its news sentiment score exceeds its recent 5-day moving average, and the number of news articles surpasses a minimum threshold (e.g., more than 5). Technical indicators like RSI and MACD are used to further confirm short-term momentum conditions.

Macroeconomic sentiment is used to adjust the portfolio's exposure to long versus short positions. For instance, when macro sentiment exceeds 0.5, the strategy allocates

more capital toward long positions. Financial statement features act as a final confirmation step to ensure that the selected stocks have adequate fundamental strength.

The portfolio is rebalanced weekly. The final stock list is determined by satisfying multiple layers of criteria, including sentiment signals, technical confirmation, and macroeconomic context.

Table 28 presents the cumulative return comparison between our strategy and the S&P 100 index from 2020 to 2025. The strategy consistently outperforms the benchmark, offering both higher long-term returns and better resilience during periods of market volatility.

These results highlight the effectiveness of combining three distinct categories of semantic features:

1. **News sentiment**, which captures short-term market reactions.
2. **Earnings outlook**, which reflects firm-level fundamentals.
3. **Macroeconomic outlook**, which provides a broader economic view.

By integrating semantic and technical indicators, the strategy is able to make dynamic, context-aware trading decisions and consistently generate alpha across a wide range of market conditions.

Table 28: Performance Comparison: Our Strategy vs. S&P 100

Metric	Our Strategy	S&P 100
Total Return (%)	153.43	90.49
Sharpe Ratio	0.725	0.731
Max Drawdown (%)	-35.94	-28.93

The table above compares the performance of the proposed strategy with the S&P 100 index. The strategy achieves a substantially higher total return (153.43% vs. 90.49%),

indicating strong cumulative growth over the evaluation period. Although its Sharpe ratio (0.725) is slightly lower than that of the benchmark (0.731), the difference is marginal, suggesting comparable risk-adjusted performance. The strategy also experiences a larger maximum drawdown (−35.94% vs. −28.93%), reflecting higher interim volatility. Overall, these results highlight that while semantic-enhanced trading signals can generate greater long-term returns, they may also involve increased exposure to downside risk.

Chapter 6. Conclusions

6.1 Research Summary

This study proposed a stock selection framework that leverages LLMs and prompt engineering to extract semantic signals from diverse financial texts. By designing task-specific prompts, we enabled LLMs to analyze three types of financial documents: news headlines for short-term market sentiment, earnings reports for firm-level fundamentals, and macroeconomic announcements for broader economic context.

The unstructured texts were processed through zero-shot or chain-of-thought prompting strategies, allowing the LLM to assign interpretable sentiment labels or polarity scores. These semantic outputs were then transformed into structured features such as daily sentiment averages, changes, moving averages, and volume measures. These features were combined with traditional technical and price-based indicators to serve as inputs for return prediction models, implemented using LSTM networks.

We evaluated the effectiveness of these semantic features through both regression-based prediction and rule-based trading strategies. Empirical results across multiple stocks and configurations demonstrated that the inclusion of LLM-derived semantic features improves prediction accuracy and provides complementary information to traditional signals. When applied in rule-based trading strategies, these features contributed to consistent and stable excess returns, outperforming the S&P 100 benchmark over the evaluation period. This validates the practical potential of incorporating language-driven signals into modern investment decision-making frameworks.

6.2 Research Contributions

This thesis contributes in several key aspects:

1. **Modular semantic extraction framework:** This study proposes a flexible and modular framework that applies zero shot prompting techniques to extract semantic signals from heterogeneous financial documents, including news headlines, earnings reports, and macroeconomic event texts. The design offers a highly scalable and flexible solution, making it suitable for diverse applications in the financial sector.
2. **Integration of CoT prompting in financial reasoning:** The study introduces the use of CoT prompting to interpret complex financial narratives. This approach improves the transparency of language model reasoning and enables interpretable sentiment extraction through structured intermediate steps.
3. **Explainable LLM Semantic Features:** This study demonstrates that semantic features generated by large language models are both interpretable and predictive. The signals extracted via prompting are not black-box outputs; rather, the LLM can articulate the reasoning behind each output in natural language, enabling sentiment evaluations and thematic interpretations to be traced back to specific textual evidence and intermediate reasoning steps. Empirical results show that these semantic features significantly enhance stock return prediction performance, and when combined with price and technical indicators, they serve as effective inputs for predictive models and rule-based trading strategies.
4. **Practical applicability through rule based trading strategies:** The study connects semantic features with real world investment decision making by designing interpretable,

rule based trading strategies. These strategies consistently outperform the S&P 100 benchmark in backtesting, highlighting the practical value of language driven insights in asset management.

6.3 Research Limitations

This study also has several limitations.

First, the effectiveness of semantic interpretation still depends on the quality of prompt design and the stability of LLM inference. Although prompt engineering reduces the need for labeled data, it introduces sensitivity to phrasing and contextual framing, which may affect consistency across different document types or market narratives.

Second, the selection and weighting of semantic features are manually defined rather than optimized through automated methods. While this simplifies interpretation, it may not yield the most predictive feature combinations and could limit the model ’ s generalizability across varying market conditions or asset classes. Future work could explore feature selection through data-driven or learning-based techniques.

Third, while semantic features are interpretable, their interaction with traditional price and technical indicators may be non-linear and regime-dependent. The dynamics between textual signals and market movements under different economic cycles, volatility regimes, or shock events have not yet been fully examined and warrant further investigation. Moreover, the current approach overlooks the varying influence of different macroeconomic indicators, treating them as equally weighted in aggregation. This simplification may obscure meaningful market signals and further limit the model ’ s interpretability and predictive performance in practical applications.

6.4 Future Works

First, one promising direction is to integrate few shot prompting or retrieval augmented generation to improve the reliability and domain specificity of LLM outputs. These approaches help reduce semantic drift and improve consistency, particularly when dealing with lengthy or technical financial documents.

Second, future research could broaden the use of semantic modeling to include a wider variety of industries and asset classes. This would test the generalizability of LLM based frameworks on heterogeneous financial texts. Additionally, using multiple data types, like merging financial text with tables or news with social sentiment, can strengthen the model's understanding of market complexity.

Lastly, reinforcement learning can be explored to dynamically optimize trading strategies based on evolving semantic signals. Compared to rule-based logic, reinforcement learning enables the model to learn adaptive decision-making strategies that respond to different market conditions and signal interactions, thereby enhancing performance in real-world applications. In the long run, LLMs may go beyond being mere analytical assistants and take on the role of decision-makers, directly driving the generation and execution of trading strategies.

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